PHARMACOKINETICS II:
METABOLISM & ELIMINATION

Dr Melissa Mohammad Hirman

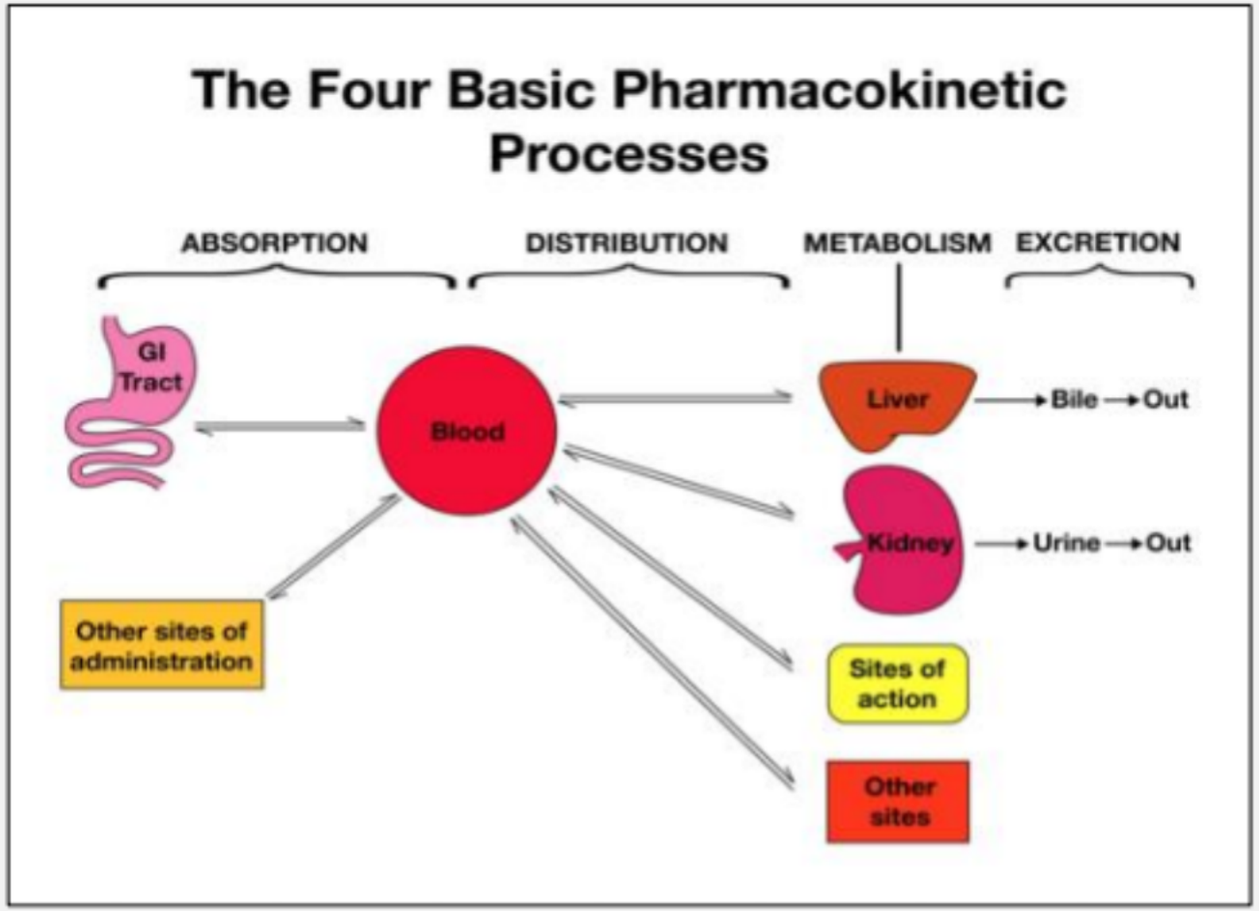
B.Pharm (Hons), M.Pharm (ClinPharm), PhD (ClinPharm)

Universiti Malaysia Sarawak

OBJECTIVES

- List the processes of termination of drug activity.
- Understand the redistribution and uptake processes.
- Discuss the enzyme induction and enzyme inhibition with examples.
- State phases and outcomes of biotransformation with examples.
- Outline the determinants of biotransformation.
- List different routes of drug excretion.
- List processes of renal excretion.
- Discuss the factors that influence biotransformation and renal excretion.
- Discuss the enterohepatic circulation.

METABOLISM

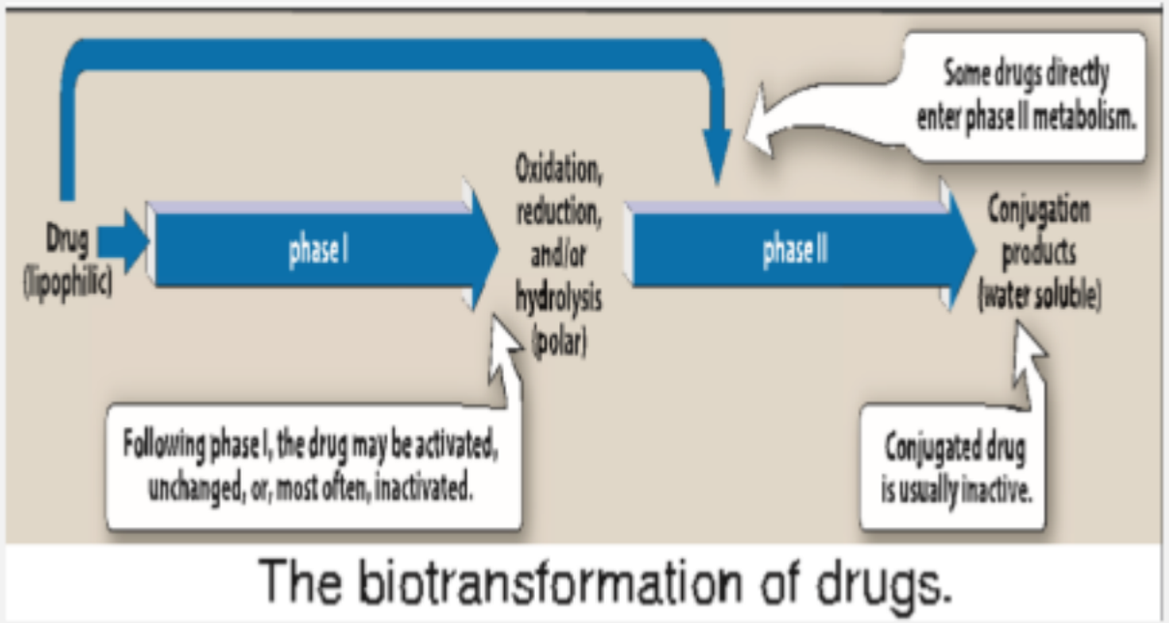

The Four Basic Pharmacokinetic Processes
ABSORPTION
DISTRIBUTION
METABOLISM
EXCRETION
GI Tract
Blood
Liver
Bile
Out
Kidney
Urine
Out
Other sites of administration
Sites of action
Other sites

METABOLISM

- Most drugs are metabolized by the liver via hepatic enzymes in phase 1 and phase 2 reactions.

- However, it still can occur in the lungs, kidneys & intestines.

- Metabolism is also called **biotransformation,** which is chemical alterations to drug molecules.



The biotransformation of drugs.

- The drug is metabolized to a /an:

✓ Inactive form

✓ Active form (applicable to prodrugs e.g. clopidogrel) prevent dangerous blood clots

✓ Water soluble form (easy to be eliminated through urine)

✓ Toxic metabolites

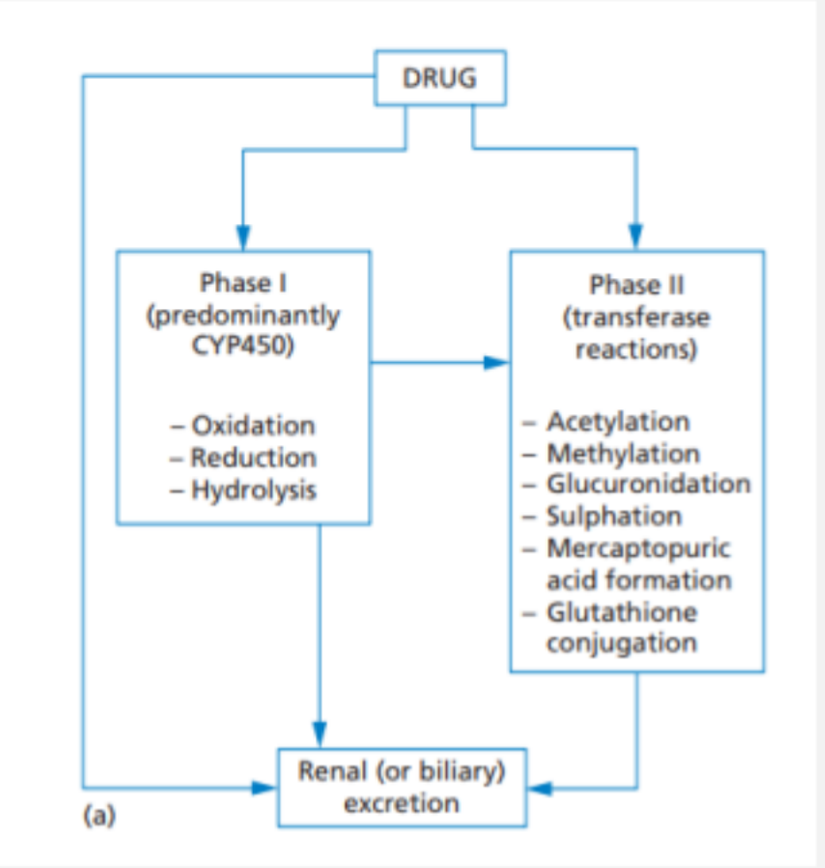
Note: The phases of biotransformation may not occur in order

METABOLISM

Phase I:

- The primary hepatic enzyme system in drug metabolism is the cytochrome P-450 enzymes (CYP-450 enzymes), also known as microsomal enzymes.

- CYP-450 system is one of the most important systems that influences **drug-drug interactions**.



METABOLISM

CYP450 Enzyme Induction & Inhibition

Enzyme Inducers:
Rifampicin
Phenytoin
Phenobarbitone
Carbamazepine

Enzyme Inhibitors:
Azoles
Omeprazole
Erythromycin
Ritonavir

conjugated drug = inactive

- Enzyme Inducers:

✓ Enzyme inducers increases the metabolism of drugs resulting in lower active drug concentration.

✓ e.g. Warfarin (blood thinner) - Rifampicin

→ when given with an enzyme inducer, therapeutic levels of warfarin drops.
→ Thus, the patient is at risk of blood clotting as the drug can't work effectively.

- Enzyme Inhibitors:

✓ Enzyme inhibitors decreases the metabolism of drugs resulting in higher active drug concentration.

✓ e.g. Warfarin (blood thinner) - Erythromycin

→ when given with an enzyme inhibitor, therapeutic levels of warfarin increases.
→ Thus, the patient is at risk of bleeding as the drug concentration increases at toxic levels.

METABOLISM

Clinical significance of enzyme induction & inhibitions:

- Drug interactions (therapeutic effects & toxicities)

❑ *Oral contraceptives – rifampicin*

- decrease level of contraceptives causing decreased effectiveness

❑ *Clopidogrel – rifampicin*

- clopidogrel is a **prodrug**
- rifampicin increase levels of clopidogrel which increases antiplatelet effect, increased bleeding!!

❑ *Clopidogrel – omeprazole*

- affects conversion of inactive form to active form, thus reducing its effectiveness

- Drug Autoinduction
- Sometimes a drug that can induce liver enzymes, can increase it's own metabolism (e.g carbamazepine, phenytoin).
- This can lead to decreased levels leading to drug tolerance.

Enzyme Inducers:
Rifampicin
Phenytoin
Phenobarbitone
Carbamazepine

Enzyme Inhibitors:
Azoles
Omeprazole
Erythromycin
Ritonavir

Therefore, it may be needed to measure the serum levels of certain drugs to ensure it's efficacy in certain patients.

METABOLISM

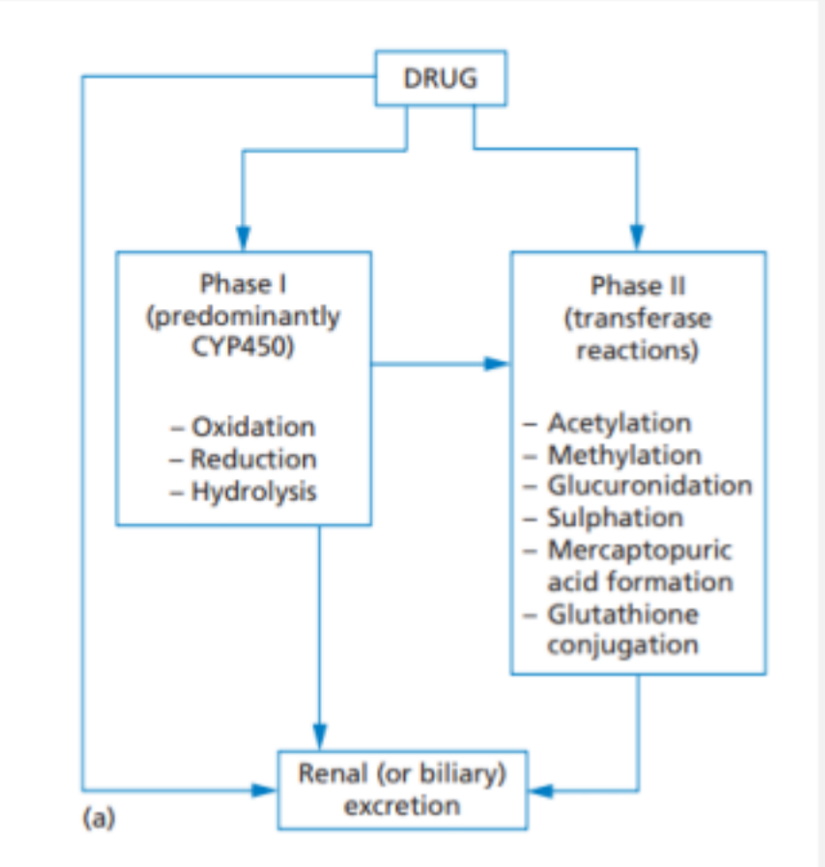
Phase 2:

- Conjugation, where the drug is metabolized into a more polar and water soluble metabolite to ease excretion.

- Involves transferases

- Reactions include:

- glucuronidation
- acetylation
- methylation
- sulphation
- glutathione conjugation



METABOLISM

Factors that affects metabolism:

- **Patient factors:**

- Liver Disease (cirrhosis etc)

→ reduces the ability to biotransform drugs. Can lead to increased active form of drug which results in toxicities

- The elderly may have diminishing liver function
- Neonates and infants liver may not be fully matured yet

- **Polymorphism** (how readily the drug is metabolized e.g. fast or slow)**:**

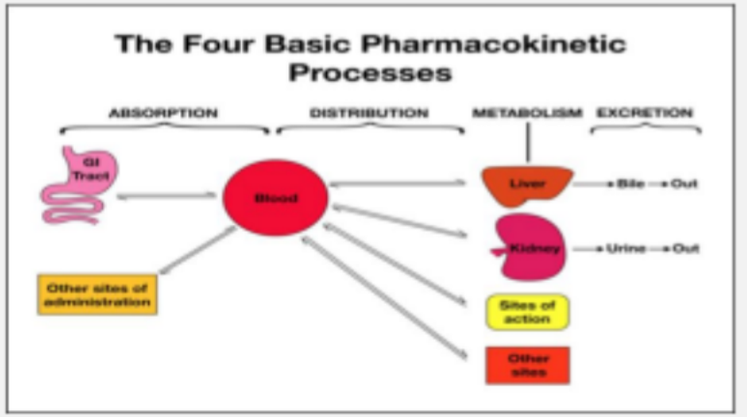
- Rapid metabolizers:

→ active drug is biotransformed rapidly to inactive form → reduces concentration of active drug → reducing it's therapeutic effect.

- Slow metabolizers:

→ active drug is biotransformed slowly to inactive form → increases the concentration of active drug → lead to toxicities.
e.g. codeine in babies (increased somnolence which can lead to respiratory depression)

EXCRETION



- Excretion or elimination of the drug from the body is through the **renal** (urine, majority) and **non-renal** routes (liver, lungs, sweat, saliva etc.)

- **Half-life , $t_{½}$**

✓**amount of time it takes for half of the medication to be eliminated in the body**

✓e.g. it takes 8 hours for 100mg of Drug A concentration to drop to 50mg. Thus half life for Drug A is 8 hours

✓it takes 4-5 half-lives to eliminate the drug from the body



EXCRETION

- The concept of half-life, $t_{1/2}$ can be used to determine the **steady state (SS) of the drug**.

- **Steady state concentration, C_{ss}**

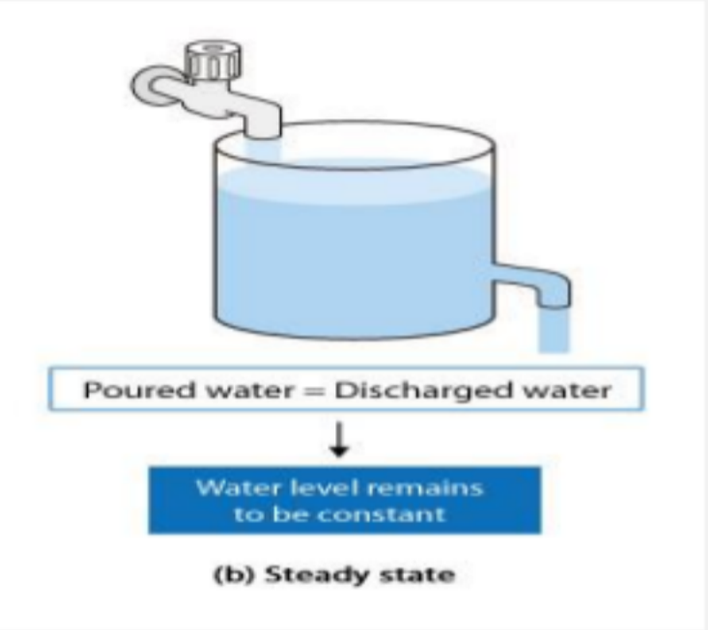
✓ **when the concentration of the drug in the body stays consistent**.

✓ rate of administration = rate of elimination

✓ occurs after four to five half-lives of administered drug

✓ concentration below SS → ineffective

✓ concentration above SS → toxic

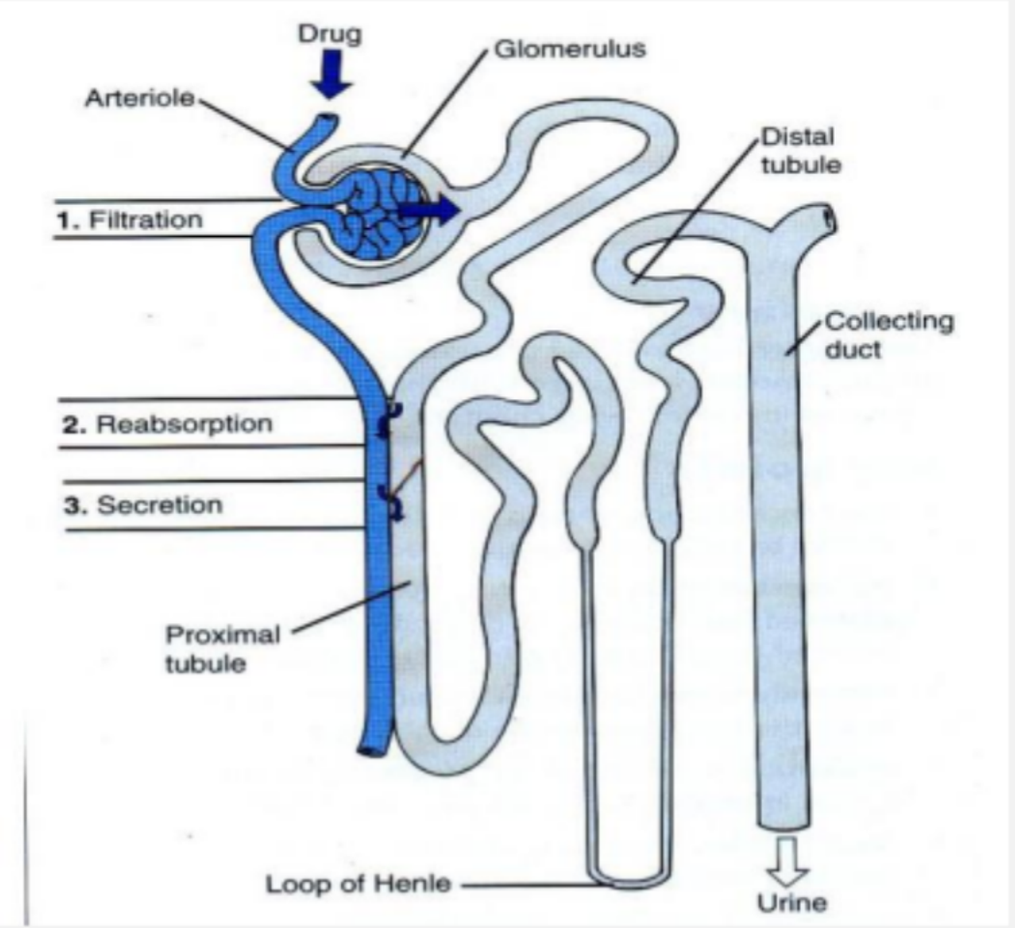


EXCRETION

RENAL EXCRETION

Processes involved:

1. Glomerular filtration (passive)
2. Tubular reabsorption
3. Tubular secretion



- Changes in kidney function can affect the elimination of drugs.
- Kidney function is measured by lab values such as serum creatinine, glomerular filtration rate (GFR), and creatinine clearance (CrCL).

EXCRETION

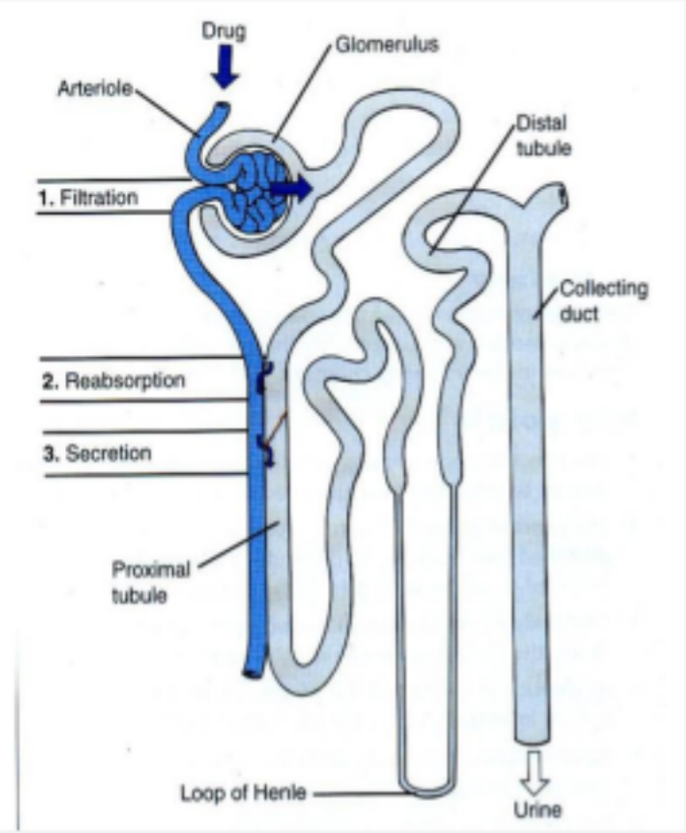
1. Glomerular filtration

- Drug is filtered through the glomerulus into the bowman capsule

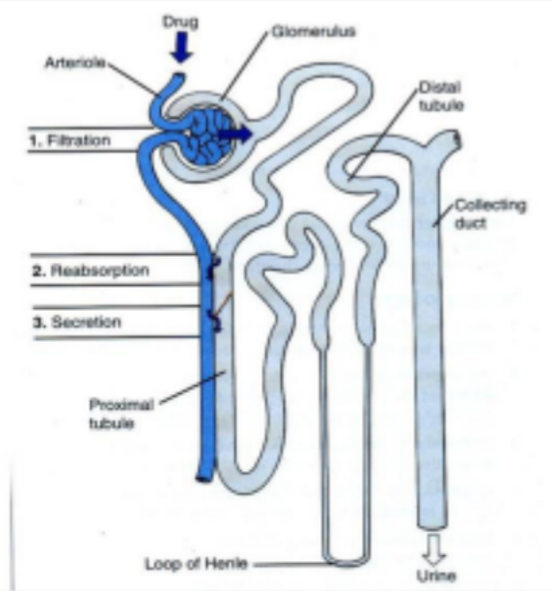
- Glomerular filtration depends on:
- Molecular weight / size of drug molecules (small molecules get filtered easily)

- **plasma protein binding** of the drug
  - Highly protein bound drugs is difficult to be filtered
  - Can increase drug concentrations leading to toxicities

- **glomerular filtration rate (GFR)**
  - patients with renal impairment / kidney disease have reduced GFR which can result accumulation of the drug.
  - Dose adjustments may be needed.



EXCRETION



2. Tubular Reabsorption

- Drugs that are small, non polar and hydrophobic are easily reabsorbed back to the circulation (passive diffusion)

- Once reabsorbed, it goes into the liver and metabolized into a more polar, water soluble form for easier excretion.

- Urine pH can affect drug reabsorption and excretion
  - as it can affect the ionization of the drug
  - Example: alkalinizing the urine will cause ionization of acidic drugs which will reduce its reabsorption and increase its excretion in the urine
  - This concept is often used in treating drug overdose

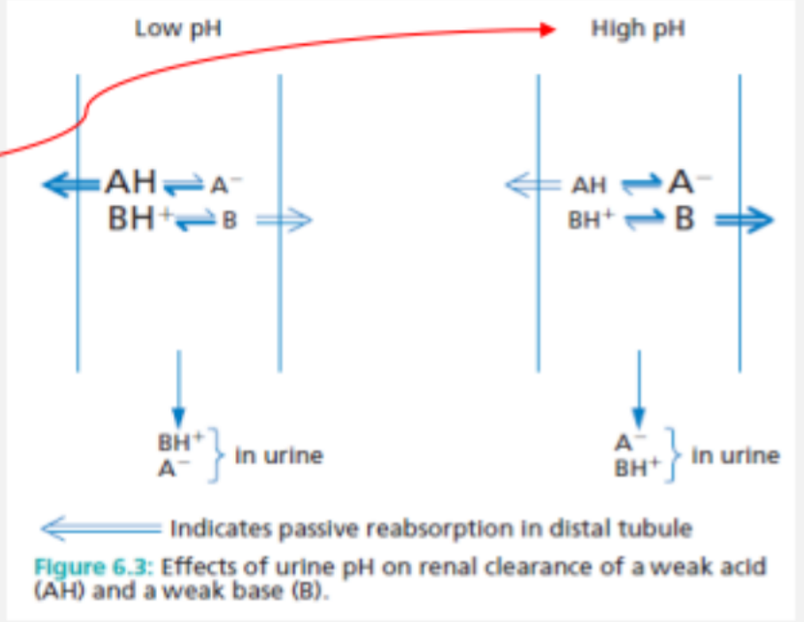


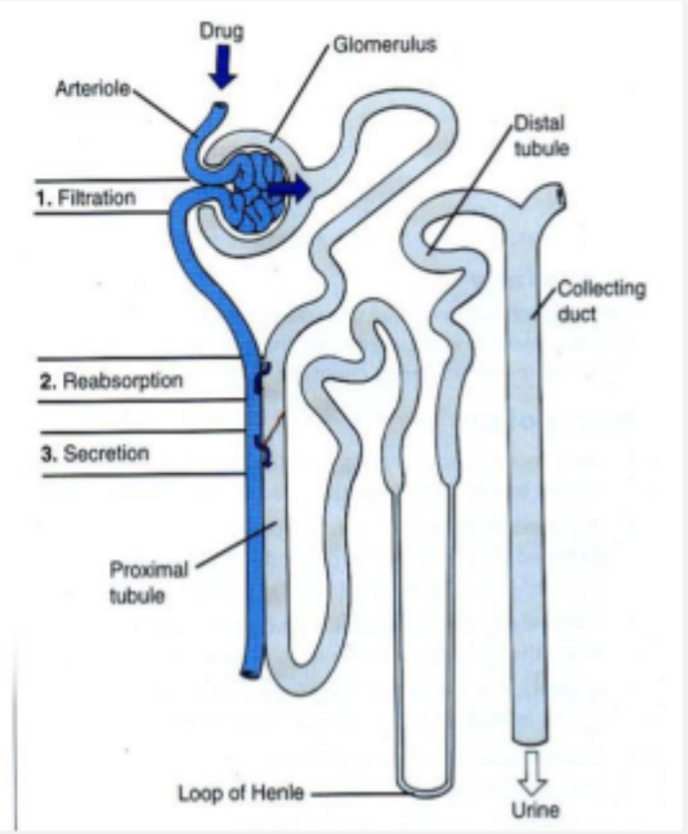
Figure 6.3: Effects of urine pH on renal clearance of a weak acid (AH) and a weak base (B).

EXCRETION

3. Tubular secretion

- Some drugs do not get filtered through the glomerulus.
- Secretion occurs at the proximal tubule
- This depends on the **solubility** of the drug and **concentration gradient across the blood and proximal tubule**
- Non-polar, hydrophobic drugs and higher concentration in blood vs lower concentration in tubules would ease secretion
- Polar, hydrophilic and lower concentration in blood vs higher concentration in tubules would need **active transportation** into the tubules

➢ This is a site potential for drug interactions through transporter competition (e.g. probenecid increases penicillin levels by competing for these transporters, cimetidine increases TMP-SMX levels)



EXCRETION

BILIARY EXCRETION

- Important route of excretion for drugs that are highly polar and high molecular weight

- **Enterohepatic Circulation**:

➢ A repeated process where the active form of the drug is released back into the circulation after metabolism

➢ Can prolong the effect of certain drugs (e.g. oral contraceptives, morphine, rifampicin)

➢ Example on oral contraceptives:

- excreted in bile as a glucuronide conjugate, bacteria-derived enzymes cleave the glucuronide so that free drug is available for reabsorption in the terminal ileum.

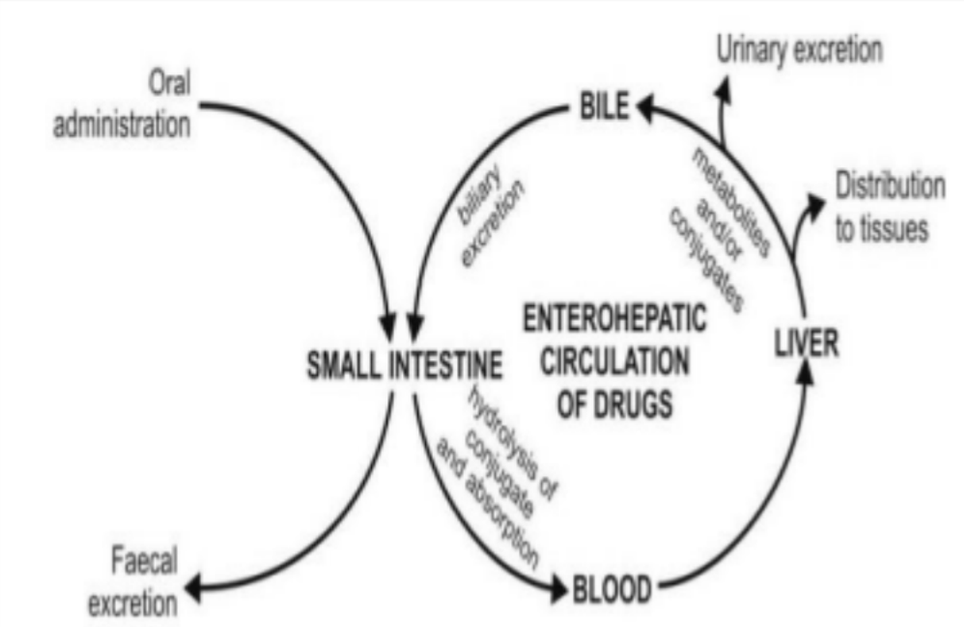


Fig. 6.5. Enterohepatic cycling of drugs

➢ Possible drug interactions can occur:

- Antibiotics can alter intestinal flora affecting the reabsorption process
- Drugs that reduce gastric motility (e.g. antidiarrheal drugs) may reduce the effect of enterohepatic cycling

LET'S RECAP… A D M E

Absorption:
- Movement of drugs across the cell membrane into the blood stream
- Bioavailability, F

- fraction of unchanged drug reaching the systemic circulation following administration by any route

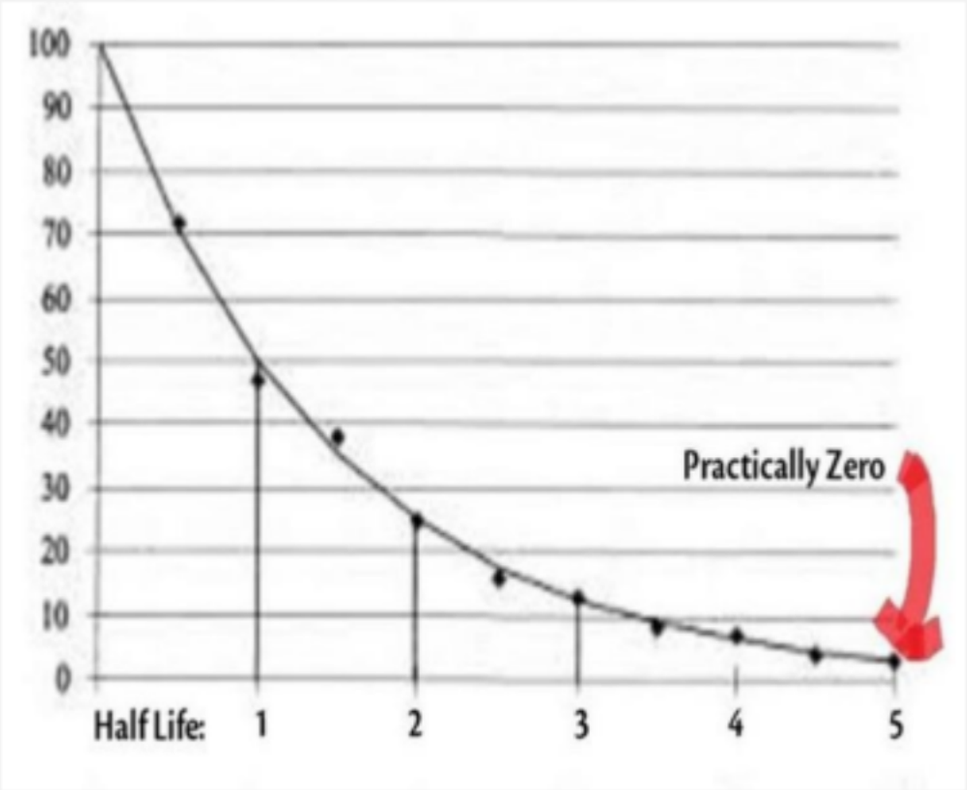
Distribution:
- movement of drugs from the circulation to the tissues and organs
- Volume of Distribution, Vd

- Volume of drug distributed to tissues and fluids

Metabolism:
- Biotransformation of a drug

Excretion:
- Elimination of a drug from the body
- Half-life, $t_{1/2}$

- amount of time it takes for half of the medication to be eliminated in the body



It takes 4-5 half-lives to eliminate the drug from the body

LET'S RECAP…

First Order Kinetics:

Drug moves at a *constant fraction*

e.g. 100mg → 50mg → 25mg → 12.5mg → etc….

[drug is eliminated at a fraction of half at each time]

Example: most drugs

Zero Order Kinetics:

Drug moves at a *constant amount*

e.g. 100mg → 80mg → 60mg → 40mg → 20mg → 0mg

[drug is eliminated at 20mg each time]

Example: controlled released drugs, depot injection / long acting injection


Thank
you!